

Machine Learning-based Approaches Comparison for Netflix/DAZN Streaming and Real Traffic Prediction

1st E. Reticcioli

DISIM

Università degli Studi dell'Aquila

L'Aquila, Italy

enrico.reticcioli@univaq.it

2nd G.D. Di Girolamo

DISIM

Università degli Studi dell'Aquila

L'Aquila, Italy

giovannidomenico.digirolamo@univaq.it

3rd C. Di Carlo

Sonicatel s.r.l.

Pescara Italy

cristhian.dicarlo@sonicatel.com

4th F. Smarra

DISIM

Università degli Studi dell'Aquila

L'Aquila, Italy

francesco.smarra@univaq.it

5th A. D'Innocenzo

DISIM

Università degli Studi dell'Aquila

L'Aquila, Italy

alessandro.dinnocenzo@univaq.it

Abstract—Traffic prediction is a fundamental topic in modern networks for performance analysis and network planning. To this aim, modeling and optimization of modern heterogeneous network infrastructures are key factors to achieve such goals, e.g. in terms of traffic flow improvement while reducing bandwidth allocation. Identifying an accurate model of a network device (e.g., a switch or a router) behavior is crucial in order to apply advanced and optimal resource allocation techniques. Such a problem is very challenging due to non-linearities and unavailability of internal variables measurements in real devices. In this respect, a promising direction is given by an appropriate integration of System Identification and Machine Learning (ML) techniques to obtain predictive models using historical data collected from the network. In this paper, a comparison of various ML-based methodologies to learn accurate models of the dynamical input-output behavior of a network switch device, obtained by appropriately combining AutoRegressive (AR) model identification with Regression Trees (RTs) and Random Forests (RFs), is provided. Furthermore, a validation over a real dataset obtained from measurements of an Italian Internet Service Provider (Sonicatel S.r.l.) is provided, as well as a comparison with classical and widely used ML methods.

Index Terms—Network Traffic Prediction, Machine Learning, System Identification, Network Optimization

I. INTRODUCTION

Communication networks involve the interconnection of a large number of devices, protocols and applications, as well as specific Quality of Service (QoS) and Quality of Experience (QoE) requirements. The heterogeneity and complexity of such infrastructures pose a number of difficult problems in terms of modeling, managing and optimizing network resources (see e.g., [1], [2], [3], [4], and references therein), making actions like network configuration, with predefined policies, or reconfiguration, to face network issues (e.g., faults, changes, etc.), very challenging.

One of the most difficult challenges to be addressed to apply optimization techniques to the management of such complex networks is to derive predictive models of the traffic behavior

that flows through a network device ([5], [6]), e.g. modeling the volume of the traffic in the form of bytes sent across the network through the nodes. These nodes can represent routers, or more generally any type of Point of Presence in which a communication interface occurs. In this scenario, computing technologies, such as graphic and tensor processing units, represent a good opportunity to implement advanced Machine Learning (ML) techniques in communication networks (see e.g., [7], [8]). In this context, ML approaches applied to communication networks have been investigated over the past years.

In this paper, we consider the problem of deriving a model to predict the traffic flow, in terms of number of packets and packet information level, over a finite time horizon. In particular, we use techniques that exploit a combination of RTs and RFs from Machine Learning and AR modeling from System Identification. Such predicting models can be particularly useful in modern applications, where network traffic packets are often encrypted to hide the number of bytes.

In this respect, some statistical characteristics that make prediction of network traffic particularly difficult is self-similarity and its highly nonlinear nature. This nonlinear nature has been shown to be insufficiently modeled by Poisson or Gaussian distributions [9]. Furthermore, from a data dependency point of view, network traffic is characterized by long-term dependencies that most statistical models fail to capture. For this reason, we decided to consider the network traffic prediction problem also as a time-series problem, to compare the methodologies mentioned above with standard ones. Time series predictions have been considered in several statistical and probabilistic approaches such as Autoregressive Integrated Moving Average (ARIMA) [10], [11], Bayesian Networks or Hidden Markov Models [12], [13], but also in the deep learning context with good success. In particular, Neural Networks (NNs) have demonstrated high accuracy in

time series forecasting [14]. The success of NNs in times series forecasting relies in their ability to capture nonlinear behavior and long range dependencies in time series data [15] than conventional linear statistical models.

For the problem of network devices control using prediction knowledge, a vast amount of research has already been done trying to exploit its advantages [16], [17], [18], [19], [20]. It is clear from all these studies, and many others, that the use of historical network data to derive accurate dynamical models of communication networks that can be directly used to setup optimal control problems, such as segment routing and/or queue management, is a challenging open problem.

Main contribution. In this paper we illustrate and experimentally validate a methodology to identify a modeling framework of a switch device located in real service provider network: such model is constructed with the intent to apply in future work Model Predictive Control (MPC) techniques for optimal resource allocation, as done in our previous work [21], [22] on a purely simulative validation setup. The computed model represents the behavior of the device, i.e. how act all the algorithms implemented in it. To the best of the authors knowledge, the state of the art still lacks of methods in this respect. In particular, none of the approaches in literature proposes a model identification technique with the aim of enabling direct application of an MPC problem using computationally efficient Quadratic Programming (QP) techniques to the input-output behavior of a real environment as the one mentioned above. On this line, the main contributions of this paper can be summarized as follows:

- 1) we adapt methodologies and theoretical results that we recently developed in [23] to combine Identification (AR identification) and Machine Learning (Regression Trees and Random Forests) in order to derive an accurate model of a switch device to predict traffic flows, using input/output historical data, over a finite horizon of arbitrary length N .

- 2) we validate the aforementioned techniques on a real experimental setup consisting of real traffic data from network measurements provided by an Italian Internet Service Provider (Sonicatel S.r.l.). Our validation tests evidence interesting results that can justify the application of our methodology to enable optimal control techniques with low computational complexity (i.e., Quadratic Programming), as done in [21]. In particular, we compare the prediction accuracy of our RT+AR- and RF+AR-based techniques with NNs and ARIMA. Results show that NNs provide a slightly better accuracy. However, from a practical point of view, such small improvement does not justify the use of NNs for MPC purposes in this context, as they require to solve a nonlinear optimization problem, while our RTs- and RFs-based methodologies allow for an efficient real-time implementation of the MPC in terms of computational complexity. We also show how the prediction accuracy obtained with the ARIMA models, which are widely used for network traffic prediction, is significantly lower than the one of our techniques.

Paper organization. The paper is organized as follows. In Section II the Sonicatel s.r.l. network is illustrated. In Section

III the model identification technique formulation is addressed. Finally, in Section IV the methodology validation is provided showing results in terms of prediction accuracy compared with different ML approaches.

II. NETWORK ENVIRONMENT

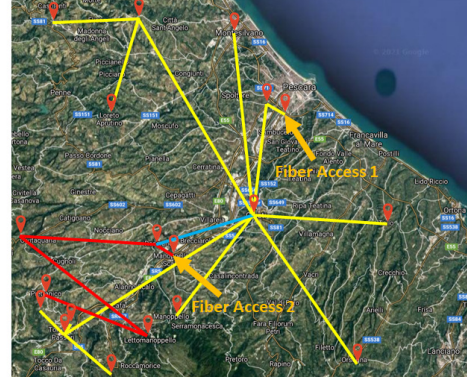


Fig. 1. Sonicatel s.r.l. network architecture in Pescara, Italy.

Sonicatel s.r.l. [24] is an Italian Internet Service Provider located in Pescara, Abruzzo. Its network is composed by two separated entities (networks), schematized in yellow and red in Figure 1. Each network has its own access to the Italian backbone fiber for internet connection. The blue link is used only in the case where the access to one of the two fiber does not work: in this case all the traffic is sent via this link to the other fiber access. Today, Sonicatel serves approximately 900 customers, 70% of which residential customers and the remaining 30% business customers. These users are equally divided between the *Fiber Access 1* of Pescara and the *Fiber Access 2* of Manoppello Scalo. To use their subscription, users must connect to the wireless access points located at the endpoints represented in red. The monitored device is a Mikrotik CCR1009-7G. It was located at a first place in *Fiber Access 1*, and later on moved for business reasons to *Fiber Access 2*. To harvest data from this device the software Cacti [25] has been used. Cacti generates a *.rrd* file that contains all the samples collected in a day. Data collection interval, which Cacti refers to as a polling interval, has been set to 5 minutes: this means a collection of 288 samples per day. A Python script has been designed to extract every day all the 288 samples stored inside the *.rrd* files to a local database. Every sample contains the timestamp, the downloaded packets from the Italian fiber backbone in the last 5 minutes (we will refer to these packets as "Received"), the uploaded packets to the same access point in the last 5 minutes (we will refer to these packets as "Transmitted"), the Voip packets generated in the last 5 minutes and, only in *Fiber Access 2*, also the Netflix and DAZN packets generated in the last 5 minutes. We remark that Voip packets also contain the services related to videoconferencing (like Skype, Teams, etc.), and as it is shown in Section IV this type of traffic has undergone a deep change with the advent of Covid19. To mark Received, Transmitted and Voip traffic with Cacti we used the digital object identifier (DOI), which is a number associated also to the producer of a network device, while to mark Netflix and DAZN packets

we used the Deep Inspection Packet (DPI). For each DPI filter inserted the Mikrotik CPU increases by about 30%, for this reason it is not possible to mark more than two types of services without losing performance on the users' QoS and QoE.

III. SWITCHED AFFINE MODELING VIA RT+AR AND RF+AR

In this section we discuss the methodology to apply the results proposed in [23], [26] to identify, starting from a set of historical traffic measurements $\mathcal{D} = \{y(k)\}_{k=0}^{\ell}$, an output only switching AR model of the traffic flow in a switch of a SDN network to predict its behavior over a fixed time horizon. Such modeling is described by the following form:

$$x(k+j+1) = A'_{\sigma_j(x(k))}x(k) + f'_{\sigma_j(x(k))}, \quad (1)$$

$j = 0, \dots, N-1$, where $x(k) \doteq [y^\top(k) \cdots y^\top(k-\delta_y)]^\top \in \mathbb{R}^{n_x}$ is an extended state to characterize a switching ARX model, with $x_\iota(k) \doteq [y_\iota(k) \cdots y_\iota(k-\delta_y)]^\top \in \mathbb{R}^{\delta_y+1}$, $\iota = 1, \dots, p$, where $p = 3$ when we use three streams of data from *Fiber Access 1* and lately from *Fiber Access 2* as described in Section II, $p = 4$ when we also use Netflix traffic packets, and $p = 5$ when we add DAZN traffic packets to the other four types of traffic. These last two cases are related only to data harvest from *Fiber Access 2*. N is the chosen future predictive horizon, and $\sigma_j : \mathbb{R}^{n_x} \rightarrow \mathcal{M} \subset \mathbb{N}$ is a switching signal that associates an operating mode in a finite set $\mathcal{M}$ to each value of $x(k)$ and each prediction step j of the horizon. **Switching AR (SAR) model identification via RT.** To derive a model as in (1), that is able to provide prediction of the traffic flow over a horizon of fixed length N we need to create pN RTs $\{\mathcal{T}_{\iota,j}\}$, $\iota = 1, \dots, p$, $j = 0, \dots, N-1$, each one to predict a specific output of the system over the N steps of the horizon. To create the tree structure (see [27]) the RT algorithm (CART) partitions the dataset $\mathcal{X}$ into regions $\mathcal{X}_i$, such that $\biguplus \mathcal{X}_i = \mathcal{X}$, and assigns to each region a constant value given by the average of the output values of the samples that ended up in that leaf. In run-time, once the trees are created, and given a real-time measurement $x(k)$ belonging to leaf i , the CART algorithm provides as a prediction the averaged value associated to the leaf. However, to obtain a dynamical modeling framework as in (5), the learning procedure needs to be slightly modified. In particular, to each leaf ι, i_j , corresponding to the partition $\mathcal{X}_{\iota,i_j}$, of each tree $\mathcal{T}_{\iota,j}$, constructed to predict the variable $y_\iota(k+j+1)$, and consequently $x_\iota(k+j+1)$, we fit the following affine model:

$$x_\iota(k+j+1) = A'_{\iota,i_j}x(k) + f'_{\iota,i_j} \quad (2)$$

where matrices A'_{ι,i_j} and f'_{ι,i_j} are as in the following Eq. (3)

$$A'_{\iota,i_j} = \begin{bmatrix} a_1 & a_2 & \cdots & a_{\delta_y} & a_{\delta_y+1} \\ 1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix}, f'_{\iota,i_j} = \begin{bmatrix} f \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad (3)$$

and its coefficients are obtained in each leaf ι, i_j by fitting the corresponding set of samples solving the following Least Squares with inequality constraints problem:

Problem 1:

$$\begin{aligned} & \underset{\xi_{\iota,i_j}}{\text{minimize}} \quad \|\Lambda_{\iota,i_j}\xi_{\iota,i_j} - \lambda_{\iota,i_j}\|_2^2 \\ & \text{subject to} \quad f_{\min} \leq f \leq f_{\max} \\ & \quad \quad \quad a_{\min} \leq a_j \leq a_{\max} \end{aligned} \quad (4)$$

where ξ_{ι,i_j} , λ_{ι,i_j} , and Λ_{ι,i_j} contain respectively the parameters of matrices in (3), the predictions $x_\iota(k+j+1)$, and the current measurements of $x(k)$. Linear inequalities (4) are used to constrain elements in ξ_{ι,i_j} to take into account physical system constraints and improve prediction accuracy. Model (2) can be easily compacted in the following form taking into account all the states $\iota = 1, \dots, p$:

$$x(k+j+1) = A_{\sigma_j(x(k))}x(k+j) + f_{\sigma_j(x(k))}, \quad (5)$$

where $\sigma_j : \mathbb{R}^{p(\delta_y+1)} \rightarrow \mathcal{M} \subset \mathbb{N}$.

SAR model identification via RF. RFs [28] basically consist on a collection of RTs, and the output prediction is given by the average of the predictions of each tree in the forest. RF-based models can be constructed upon the RT-based formulation described above. In particular, let us consider pN RFs $\mathcal{F}_{\iota,j}$, $\iota = 1, \dots, p$, $j = 0, \dots, N-1$. For each tree $\mathcal{T}_{\tau,\iota,j}$ of the forest $\mathcal{F}_{\iota,j}$, it is possible to estimate the coefficients a_* and f in (3) for each leaf ι, j, i_τ , i.e. $\tilde{\xi}_{\iota,j,i_\tau}$, solving Problem 1. With a small abuse of notation, we indicate by $|\mathcal{F}_{\iota,j}|$ the number of trees in the forest ι, j . Then $\forall \iota, j$, the parameters to build matrices in (2) can be obtained by averaging parameters a_* and f , $\forall \tau = 1, \dots, |\mathcal{F}_{\iota,j}|$, i.e.

$$\tilde{\xi}_{\iota,j} = \frac{\sum_{\tau=1}^{|\mathcal{F}_{\iota,j}|} \tilde{\xi}_{\iota,j,i_\tau}}{|\mathcal{F}_{\iota,j}|}, \quad (6)$$

over all the trees of forest $\mathcal{F}_{\iota,j}$.

IV. RESULTS

In this section we compare the accuracy obtained by the RTs- and RFs-based techniques with respect to predictions obtained using NNs and ARIMA models. We use a dataset measured from a real network device (Mikrotik CCR1009) of an Italian internet provider (Sonicatel S.r.l.). We recall that a neural network is a collection of algorithms that aims to identify the underlying relations in a dataset: it consists of groups of connected neurons organized in layers, where the connections between neurons are modeled using weights. The produced signal is then sent as input into an activation function which is in general nonlinear. The reader is referred to [29] for more details. In this work, we make use of the Deep Learning Toolbox of Matlab to compute the NNs-based models. Due to some constraints in data collection policies, the provider was able to only measure the sum of all packets entering and leaving the device, so it was possible to extract from such traffic the incoming VOIP packets only. This is because such device, located in this case in *Fiber Access 1*, is part of a running commercial network. Only after a while, when the device under consideration was moved to *Fiber Access 2* for business reasons, we were able to also measure Netflix traffic, and only recently DAZN. Thus, for this last two streaming providers we have a shorter dataset.

A. Comparison between NN, RT+AR and RF+AR

As usual, we split the collected dataset into a train dataset, used to identify the models, and a test dataset, used to validate and compare them. The datasets are composed by samples collected with a sampling time of 5 minutes, as already mentioned. Each sample is composed by the current measurement of the output $y(k)$, 3 to 5 different packet patterns, together with the regressive terms $y(k-1), \dots, y(k-\delta_y)$. The most significant identification parameters we tuned to learn the models are summarized in Table I. The predictive horizon has

TABLE I
IDENTIFICATION PARAMETERS.

RT	Value	RF	Value	NN	Value
N	10	N	10	N	10
δ_y	5	δ_y	51	δ_y	51
Minleaf	2	Minleaf	5	#layer	2
		#trees	30	#neurons	10-20

been set to $N = 10$, i.e 50 minutes, considering that larger values slightly improve the control performance at the price of much more computational complexity. The regressive index δ_y and the minimum number of samples we want in each tree of each forest (MinLeaf) have been chosen with a trial and error approach, considering that a small regressive index and a very large values for MinLeaf may lead to inaccurate predictions (as they do not provide sufficient information on the past); while a very large regressive index and a very small value for MinLeaf may also lead to inaccurate predictions (as they interpolate very old data that might negatively affect the results and produce overfitting). For the RFs-based method, according to the Out-of-Bag error of each model, we found a good compromise between complexity and quality of the model, training 30 trees for each forest. Regarding specific parameters

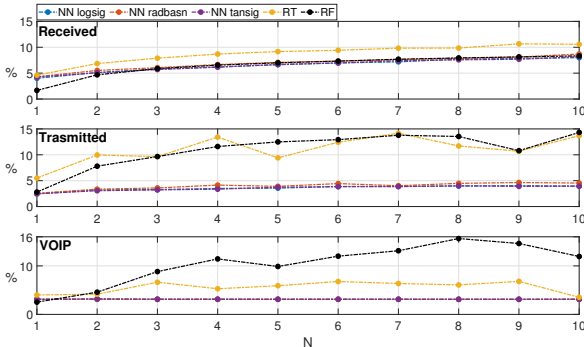


Fig. 2. NRMSE of the packets predictive models over a time horizon of $N = 10$, *Trainset* = 26138samples ~ 91 days, *Testset* = 19144samples ~ 66 days.

used for running NN, and for the sake of a fair comparison, we tuned them to obtain the best performance: we considered shallow networks of 2 layers with 10 neurons, since larger numbers did not improve the accuracy and, instead, had the negative effect of increasing the sensitivity with respect to the initial conditions of the weights. Among the numerous algorithms for optimizing neuron weights, we exploited the *Scaled conjugate gradient back-propagation* described in [30], which

provided the best accuracy over our dataset. As activation functions, we considered both the classical sigmoid function (*logsig*), the Hyperbolic tangent sigmoid transfer function (*tansig*) and also the Radial basis normalized transfer function (*radbasn*). To compare the prediction accuracy we considered the Normalized Root Mean Square Errors (NRMSE). We compare in Figure 2 the NRMSE of the different identification approaches for each packets class and over a horizon that ranges from $N = 1$ to $N = 10$. For this comparison we used a train dataset of 26138 samples, i.e. ~ 91 days, and a very large test dataset of 19144 samples, i.e. ~ 66 days, both harvested in the *Fiber Access 1* position.

Results discussion. Results emphasize that RTs, even though very simple and fast to compute, usually provide the worst accuracy: they are always worse with respect to NNs, and worse two over three times with respect to RFs. Moreover, they are always worse than all the other methods for $N = 1$, that is the most important step in predictive control strategies. NNs provide lower performance in 2 cases over 3 for $N = 1$, while they behave better over the long-term horizon. After changing

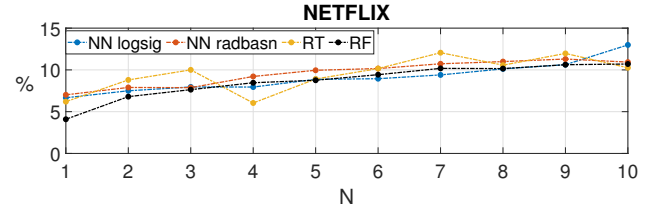


Fig. 3. NRMSE of the packets predictive models over a time horizon of $N = 10$, *Trainset* = 15744samples ~ 55 days, *Testset* = 600samples ~ 2 days.

the position of the device in *Fiber Access 2*, as explained in Section II, we were also able to track the packets coming from Netflix. We applied the same learning algorithms with a train dataset of 15744 samples, ~ 55 days, and a test dataset of 600 samples, ~ 2 days. We used the same parameters for RTs and RFs, while for the NNs we had to increase the neurons number to 20, because with only 10 neurons it was impossible to get good predictions, with the drawback of greatly increasing both the computation time and complexity of the model. Even in this case, by looking at the prediction accuracy in terms of NRMSE in Figure 3, we can see that RFs-based model perform better than NNs and RTs for $N = 1$ and worse in some cases over the horizon. Moreover, we did not include the Hyperbolic tangent sigmoid transfer function (*TanSig*) as we have not been able to obtain reasonable predictions. Finally, we analyzed the DAZN packets. The traffic produced by this provider represents a challenge from the point of view of prediction, as most of it is produced by the streaming of sporting events which are mostly concentrated in the evening hours of the weekends, but not exclusively. We created the test set using the data of a week-end (2 days), when traffic peaks are higher and more frequent, and thus when the prediction is therefore more complex. Figure 4 shows the accuracy of the prediction results over an horizon of 10 step. For the DAZN traffic we can see how the RFs perform better than all other models. Furthermore, with the addition of this information, and

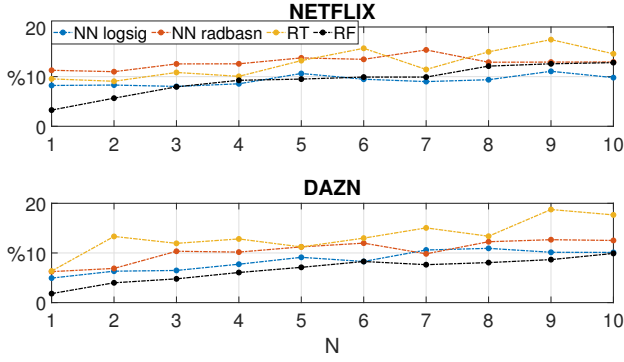


Fig. 4. NRMSE of the packets predictive models over a time horizon of $N = 10$, $\text{Trainset} = 8291\text{samples} \sim 29\text{days}$, $\text{Testset} = 600\text{samples} \sim 2\text{days}$.

for this test dataset, we were able to obtain better predictions on all other types of traffic.

Conclusions. For Transmitted packets, even if NNs perform better in the first two cases, we need to emphasize that their predictive model is based on nonlinear functions. This makes the derived model impractical for real-time control, as the corresponding MPC formulation turns into a nonlinear optimization problem. An additional problem when applying NN for MPC purposes is that obtaining a closed mathematical form of the nonlinear function of a Neural Network starting from neurons and weighs is not always an easy task, due to the highly nonlinear interconnections between the different layers. Analyzing the dataset distribution, we noticed in the last case a regular grid pattern that can be approximated by hyper-rectangles: since RTs and RFs base their prediction on hyper-rectangular dataset partitions, the better performance with respect to NNs is reasonable. For all these reasons we decided to carry our investigation on only with the RFs-based models, which provide the best trade off between accuracy and computational complexity in view of using the prediction in a MPC setup. We further remark that RTs, RFs and NNs identification processes rely on efficient algorithms, and computation times can be reduced by exploiting parallel computation or GPUs. Also, note that the learning procedures all run offline, therefore the computation time is irrelevant and the main issue is the model complexity (linear vs. nonlinear) of the derived model for the implementation of real-time optimal control.

B. Comparison between ARIMA and RF+AR

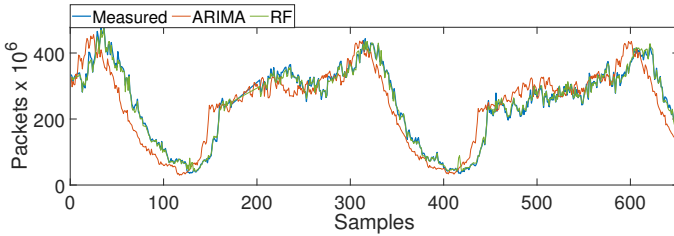


Fig. 5. Comparison between ARIMA prediction and Random Forest forecast for Received packets.

ARIMA is a model used in statistics and econometrics to measure events that happen over a period of time with

periodicity. The model is used to understand past data or predict future data in a series. It is widely used when a metric is recorded in regular intervals, from fractions of a second to daily, weekly or monthly periods. Even if ARIMA is recommended for network traffic prediction, it is necessary to emphasize that this approach needs a dataset with very limited or absent missing data, and this constraint is not always achievable in a real network. Also, ARIMA just identifies a pattern of the chosen window size inside the training dataset and uses this pattern for the predictions. In our case we used a window of $288\text{samples} = 1\text{day}$, and as shown in Figure 5 we obtained a quite high NRMSE for the Received packets equal to 10,7%, against the 2% for the RF+AR model. For the Transmitted packets we have an NRMSE of 18,2% for ARIMA against 3,6% for RF+AR. Only for the Voip packets prediction we obtain a reasonable NRMSE, about 6,46%, but the ARIMA prediction pattern is completely wrong, against 5% for RF+AR. Thus, the proposed model is confirmed as the best method for the real network traffic predictions.

C. Updating RF+AR model

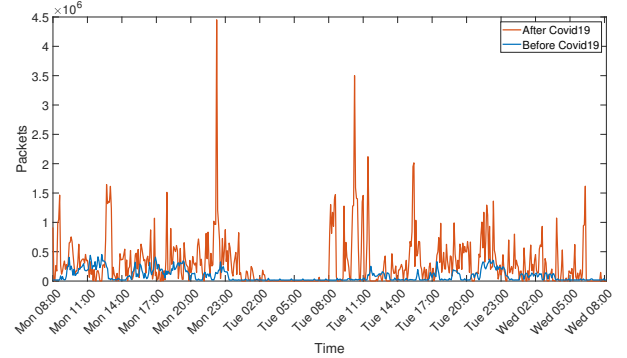


Fig. 6. Comparison, over two days, between the Voip packets recorded before the first lockdown in Italy (March 2020) with the packets of the same service after few months of the same year.

Given that the previous considerations are based on prediction methods applied to data coming from a real network, it is important to underline that the network can significantly changes over time, both due to the increase in users connected to it, and to the type of traffic required. A striking example is the change in Voip traffic with the advent of Covid19, as shown in Figure 6. These changes, if not taken into consideration with iterative updates of the predictive models, could cause future disservices for users and therefore losses for the companies themselves. With the replacement of the device in the Fiber Access 2 position, we had the opportunity to test how accurate the calculated device model would have been, in a position where the traffic pattern would have been certainly different from the traffic used for training the model. In Figure 7 is shown the NRMSE at $N = 1$ for different RF+AR models. At Update 0 we use a dataset of $45282\text{samples} \sim 157\text{days}$ harvested in *Fiber Access 1*, and than every day we compute a new model by adding 288samples harvested in the new position. The test dataset is made by $846\text{samples} \sim 3\text{days}$ and is the same for all the updates. At Update 0 we already

have very good results in terms of prediction accuracy, but is possible to see the benefit of the updates even from the first one. This result shows how it is possible to generate a model of a network device regardless of the network in which the device will be inserted without risking to degrade the QoS, and then obtain a precise model with a few days of updates. In our case after only four days we measure a saturation of the prediction accuracy improvement.

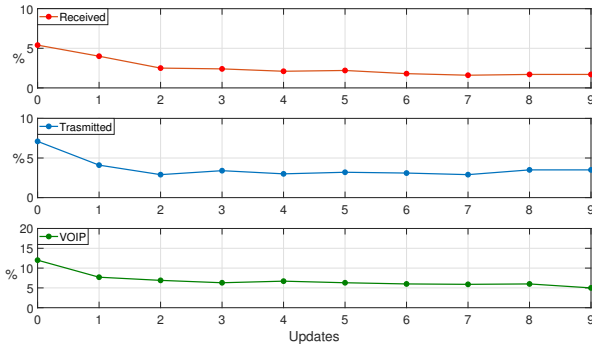


Fig. 7. NRMSE of the packets predictive models at $N = 1$ respect to model updates. Each update is $288\text{samples} = 1\text{day}$. $\text{Trainset}(0) = 45282\text{samples} \sim 157\text{days}$, $\text{Testset} = 846\text{samples} \sim 3\text{days}$.

V. CONCLUSIONS

In this paper, a procedure to derive accurate predictive models of the traffic flow has been proposed and experimentally validated over real data measured from the network of an Italian internet service provider. Results have been shown in terms of prediction accuracy both of the Received/Transmitted packets behavior and specific service packets of a network device monitored using Cacti, and show the improvements of RF+AR method with respect to some state-of-the-art prediction techniques. In future work we plan to implement optimal control strategies, leveraging the developed models, over devices located inside a real network.

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